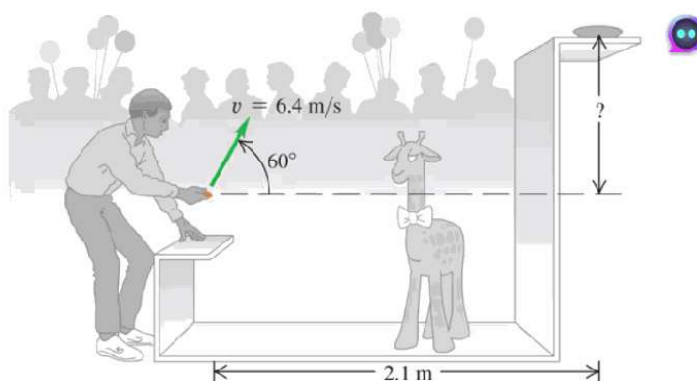


□ 1.1 Projectile motion: coin into a dish

Start from Newton's laws and explain your reasoning clearly. Determine the shelf height and the coin's vertical velocity just before landing.

- 3.19** •• **Win the Prize.** In a carnival booth, you can win a stuffed giraffe if you toss a quarter into a small dish. The dish is on a shelf above the point where the quarter leaves your hand and is a horizontal distance of 2.1 m from this point (Fig. E3.19□). If you toss the coin with a velocity of 6.4 m/s at an angle of 60° above the horizontal, the coin will land in the dish. Ignore air resistance. (a) What is the height of the shelf above the point where the quarter leaves your hand? (b) What is the vertical component of the velocity of the quarter just before it lands in the dish?

Figure E3.19



Problem 3.19, supplied semester standards